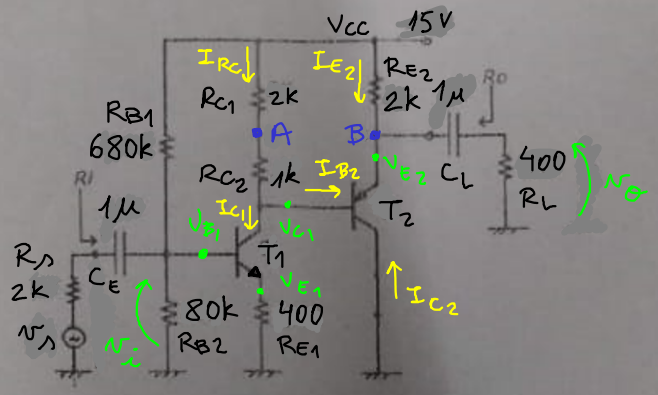
2.- $\beta = 400$; $r_x = 0$ y $V_A \to \infty$; $f_T = 200$ MHz; $C_\mu = 1$ pF)

a) Hallar la tensión de reposo de c/u de los terminales de los transistores contra común y los valores *por inspección* de Ri, Ro, A_v y A_{vs} a frecuencias medias.

b) Obtener las frecuencias de corte inferior y superior aproximadas para A_{vs}. En base a los valores obtenidos, trazar el diagrama de Bode aproximado de módulo y argumento.

c) Analizar *cualitativamente* cómo se modificarán los valores de reposo y señal si se conecta un capacitor de 10μF entre los puntos A y B.

d) Idem **c)**, si se reemplaza T_1 por un NMOSFET canal inducido en igual configuración.



$$V_{BB} = \frac{V_{CC}\, R_{B2}}{R_{B1} + R_{B2}} = \frac{15\,V \cdot 80\,k\Omega}{680k\Omega + 80k\Omega} = \frac{30}{19}\,V = 1,58\,V$$

$$R_B = R_{B1} /\!/ R_{B2} = \frac{680k\Omega \cdot 80k\Omega}{680k\Omega + 80k\Omega} = 71,58k\Omega$$

$$I_{C_1} = \frac{V_{BB} - V_{BE(ON)}}{\frac{R_B}{\beta} + R_E} = \frac{167}{110010}\,A = 1,52\,mA$$

$$V_{B1} = V_{BB} - I_B R_B = 1,89\,V - \frac{1,52\,mA}{400} \cdot 71,6k\Omega = 1,62\,V$$

$$I_{B1} = \frac{I_{C1}}{\beta} = 3,80\,\mu A$$

$$V_{E_1} = V_{B_1} - V_{BE(ON)} = 1,62\,V - 0,7\,V = 918\,mV \quad \text{NO}$$

$$V_{E1} = I_C R_{E_1} = 1,52\,mA \cdot 400\Omega = 608\,mV$$

Tomando $I_B << I_{RC} \Rightarrow I_{C_1} \approx I_{RC}$

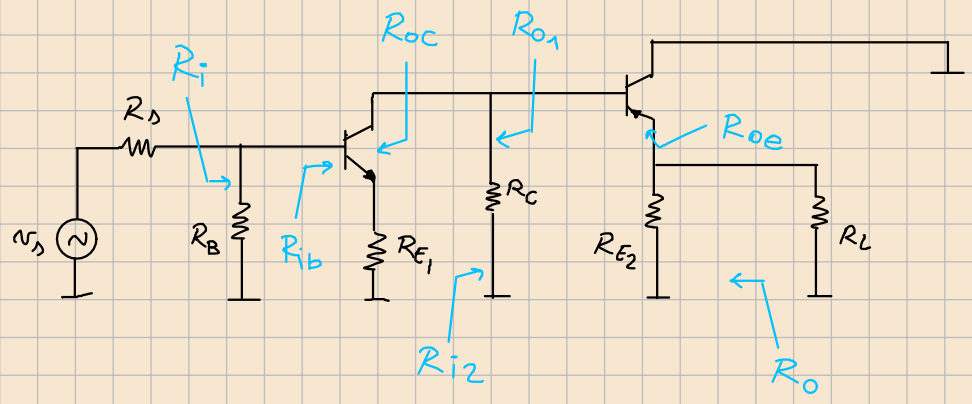
$$\Rightarrow V_{C1} \approx V_{CC} - I_{RC} R_C = 15\,V - 1,52\,mA\,(2k\Omega + 1k\Omega) = 10,44\,V$$

$$V_{E2} \approx V_{C1} - V_{BE(ON)} = 10,4\,V + 0,7\,V = 11,1\,V$$

$$I_{E2} \approx \frac{V_{CC} - V_{E2}}{R_{E_2}} = \frac{15\,V - 11,1\,V}{2k\Omega} = \frac{193}{100k}\,A = 1,93\,mA$$

T	g_m [mS]	r_d [Ω]	r_π [kΩ]	C_μ [pF]	C_π [pF]
1	59	17	6,8	2	45
2	75	13	5,3	2	57

Circuito a frecuencias medias



$$R_{oc} = r_o\,(1 + g_{m_1} R_{E_1}) \longrightarrow \infty$$

$$R_{oe} = \frac{r_{\pi 2} + R_{o1}}{\beta_2} \cong \frac{r_{\pi 2} + R_C}{\beta} = \frac{5k3\Omega + 3k\Omega}{400} = 21\,\Omega$$

$$R_o = R_{E_2} \,//\, R_{oe} = 2k\Omega \,//\, 21\Omega = 21\,\Omega$$

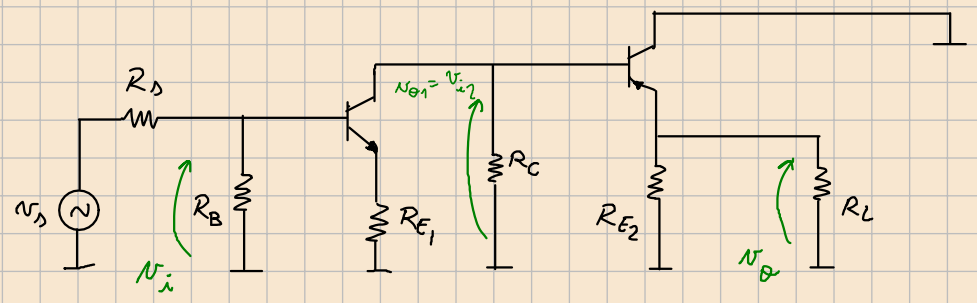
$$R_{ib} = r_{\pi 1} + \beta_1 R_{E_1} = 6k8\Omega + 400 \cdot 400\Omega = 167\,k\Omega$$

$$R_i = R_B \,//\, R_{ib} = 71,6\,k\Omega \,//\, 167\,k\Omega = 50\,k\Omega$$

$$R_{i2} = R_C \,//\, \underbrace{\left(r_{\pi 2} + \beta\, R_{ce}\right)}_{333\Omega} = 3k\Omega \,//\, \left(5k3\Omega + 400 \cdot 2k\Omega // 400\Omega\right)$$

$$R_{i2} \cong R_C = 3k\Omega$$

ganancias ↘



$R_{CE} = R_{E_2} // R_L$

$R_{CE} = 333\,\Omega$

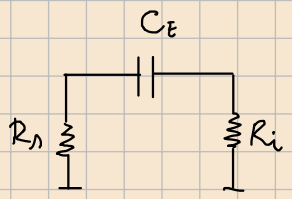
$$A_{v_2} = \frac{v_o}{v_{i2}} = \frac{g_{m_2} v_{be_2} R_{CE}}{g_{m_2} v_{be_2} R_{CE} + v_{be_2}} = \frac{75\,mS \cdot 333\,\Omega}{75mS \cdot 333\,\Omega + 1} = 0,96$$

$$A_{v_1} = \frac{v_{o1}}{v_i} = \frac{-g_{m_1} v_{be_1} R_{i2}}{g_{m_1} v_{be_1} R_{E_1} + v_{be_1}} = \frac{-59mS \cdot 3\,k\Omega}{59mS \cdot 400 + 1} = -7,2$$

$$A_v = A_{v_1} \cdot A_{v_2} = -6,92$$

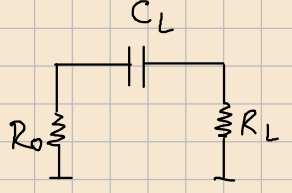
$$A_{v_S} = A_v \cdot \frac{R_i}{R_i + R_S} = -6,65$$

f_L:



$$\tau_E = C_E (R_S + R_i) = 1\mu F \cdot (2k\Omega + 50k\Omega) = 52\,ms$$

$$f_{L_E} = \frac{1}{2\pi \tau_E} = 3\,Hz$$

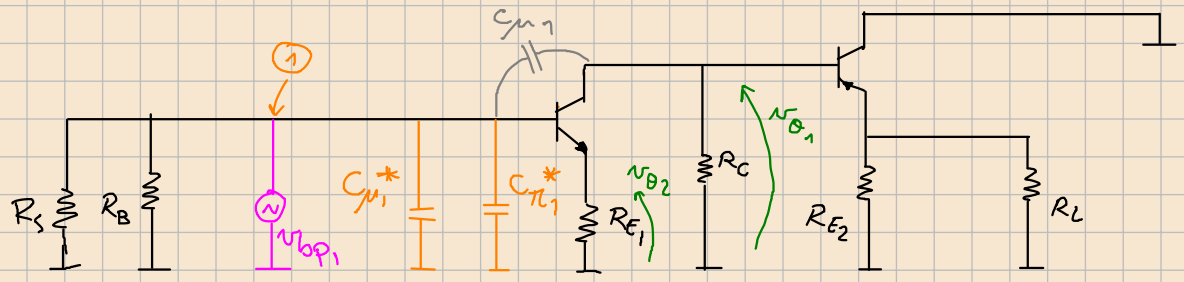


$$\tau_L = C_L (R_L + R_o) = 1\mu F \cdot (400\,\Omega + 21\,\Omega) = 421\,\mu s$$

$$f_{L_L} = \frac{1}{2\pi \tau} = 378\,Hz$$

$\Rightarrow$ $f_L \approx 380\,Hz$.

f_H



$$I = \frac{V_1 - V_2}{Z} \longrightarrow I = \frac{V_1}{Z^*} \quad , \quad Z = \frac{1}{sC} \text{ o } \frac{1}{j\omega C}$$

$$Z^* = \frac{Z V_1}{V_1 - V_2} = Z \left(\frac{1}{1 - V_2/V_1} \right)$$

$\curvearrowright^{-1}$

$$C^* = C \left(1 - V_2/V_1 \right)$$

$$C_{\mu 1}^* = C_{\mu 1} \left(1 - \frac{v_{a2}}{v_{bp1}} \right)$$

$$\frac{v_{a2}}{v_{bp1}} = \frac{- g_{m1} v_{be1} R_{i2}}{g_{m1} v_{be1} R_{E1} + v_{be1}} = \frac{- 59\,mS \cdot 3\,k\Omega}{59\,mS \cdot 400\,\Omega + 1} = -7,12$$

$$C_{\mu 1}^* = 2\,pF \left(1 + 7,12 \right) = 16,4\,pF$$

$$C_{\pi 1}^* = C_{\pi 1} \left(1 - \frac{v_{a2}}{v_{bp1}} \right)$$

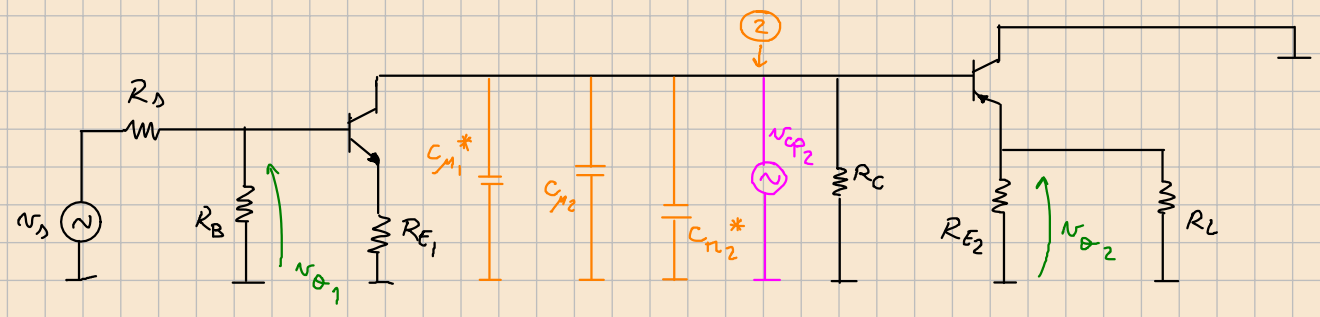
$$\frac{v_{a2}}{v_{bp1}} = \frac{g_{m1} R_{E1}}{g_{m1} R_{E1} + 1} = \frac{59\,mS \cdot 400}{59\,mS \cdot 400 + 1} = 0,96$$

$$C_{\pi 1}^* = 45\,pF \left(1 - 0,96 \right) = 1,83\,pF$$

$R_{Th} = R_S // R_B // R_{ib} \approx R_S = 2k\Omega$, $C_{Th} = 18,5\,pF$

$\tau_1 = C_{Th} R_{Th} = 37\,ns$

$f_{H1} = 4,3\,MHz$



$$C_{\mu_1}^* = C_{\mu_1}\left(1 - \frac{v_{\theta 1}}{v_{cp2}}\right) \simeq C_{\mu_1} = 2pF$$

$$C_{\mu 2} = 2pF$$

$$C_{\pi 2}^* = C_{\pi 2}\left(1 - \frac{v_{\theta 2}}{v_{cp2}}\right)$$

$$\frac{v_{\theta 2}}{v_{cp2}} = \frac{g_{m2}\, v_{be2}\, R_{E2}/\!/R_L}{g_{m2}\, v_{be2}\, R_{E2}/\!/R_L + 1} = 0,96$$

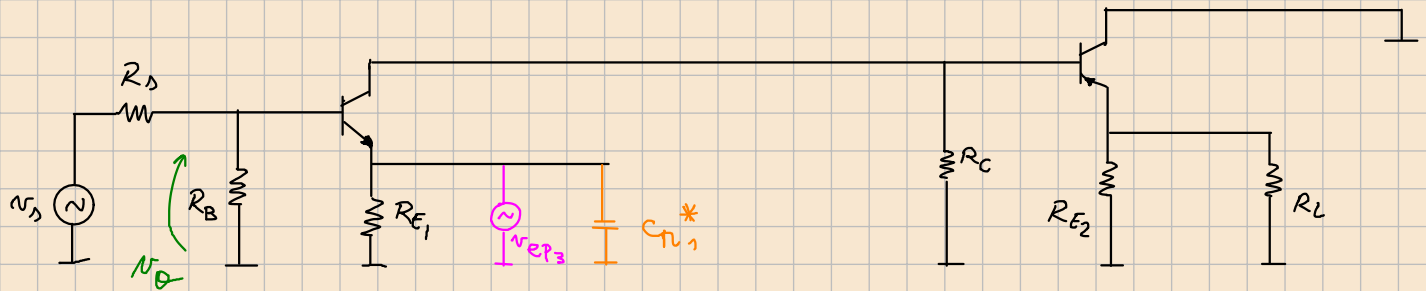
$$C_{\pi 2}^* = 57\,pF\,(1 - 0,96) = 2,3\,pF \quad \rightarrow \quad C_{Th} = 6,3\,pF$$

$$R_{Th} = R_C /\!/ R_{ib_2} \simeq R_C = 3k\Omega$$

$$\tau_2 = C_{Th} R_{Th} = 6,3pF \cdot 3k\Omega \simeq 19\,ns$$

$$f_{H2} = \frac{1}{2\pi \tau_2} = 8,4\ MHz$$

$$f_{H_{TOTAL}} = \frac{1}{2\pi(\tau_1 + \tau_2)} \simeq 3\,MHz$$



$$C_{\pi_1}^* = C_{\pi_1}\left(1 - \frac{v_\theta}{v_{ep3}}\right)$$

$$\frac{v_o}{v_{ep3}} = \frac{R_S // R_B}{R_S // R_B + r_{\pi_1}} \approx \frac{2k\Omega}{2k\Omega + 6k8\Omega} = 0,23$$

$$C_{\pi_1}^{*} = 45\,pF \cdot (1 - 0,23) = 35\,pF$$

$$R_{Th} = \frac{r_{\pi_1} + R_S // R_B}{\beta} // R_E \approx \frac{r_{\pi_1} + R_S}{\beta} = \frac{8800}{400} = 22\,\Omega$$

$$\tau_3 = 35\,pF \cdot 22\,\Omega = 770\,ps$$

$$f_{H_3} = \frac{1}{2\pi\tau_3} \approx f_T = 200\,MHz$$

